# BEGIN YOUR QUEST

healthwizkingom.github.io/Healthwiz-Kingdom

1. **Scan** or open the link: any modern browser, nothing to install.
2. Create your traveller in the **Traveller's Registry**, then log one thing.
3. Tap **Install app** or **Add to Home Screen** to play offline.


**GROUP 14**
Kolej MARA Kulim

| | |
|---|---|
| RAYYAN RAFIQIN | QAYYUM |
| NAUFAL | AMIR |
| MUI'IZZUDDIN | MUIZ ZIKRI |


**Your data stays on your device.** A free account is optional, for backup and sync only.

**Wellness tracker, not a medical device.** HealthWiz does not diagnose anything and gives no medical advice. If you feel unwell, talk to a parent, teacher or healthcare professional. In an emergency call **999** (Malaysia).

# THE QUEST

**Health tracking is easy to start and hard to keep up. In HealthWiz Kingdom every healthy action has a visible effect.**

You log **water, meals, sleep, stairs** (with workout heart rate), **runs** and **stress**. **Medius, the Wizard King,** guides you as your real actions restore the kingdom, with **Luna,** an apprentice witch, as your everyday companion.

## HOW IT WORKS

1. **Log** a real habit: a glass of water, a meal, a climb.
2. **Restore** the kingdom: its regions heal as you keep going.
3. **Understand:** a "?" shows the formula and sources behind every number.

## WHAT SETS IT APART

1. **Health first, game second.** Game values (XP, levels, badges) are tagged "game progress, not a health measurement".
2. **Every number shows its working:** inputs, formula, your numbers, sources, limits.
3. **Honest about uncertainty.** Values are labelled **measured, calculated, estimated, simulated** or **missing**. No invented readings.
4. **Private and optional.** Works on your device with no account; an account is only for backup and sync.

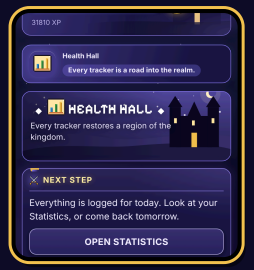
| 8 | 8 | 104 | 102 |
|---|---|---|---|
| REGIONS | THEMES | MODULES | TESTS |

# SEE IT RUNNING

REAL SCREENS FROM THE LIVE APP




**HOME**
Today's health at a glance




**HEALTH HALL**
Every tracker, one next step




**THE KINGDOM**
Eight regions, all restored




**CAMPUS MAP**
3D, with a heart on each staircase




**NUTRITION**
kcal and mL against your goal




**STRESS QUEST**
The Storm Within




**TRAINING HALL**
Pick a goal, muscle and move




**SCIENCE LAB**
Test a guess with your own logs

Captured from the running app on a phone-sized screen with 30 days of sample entries. The numbers are demo data, not real health measurements.




# Turn your health into an adventure.

A pixel-art fantasy RPG that is also a health tracker, built on real science and honest numbers.

Water · Food · Sleep · Stress · Stairs · Running · Heart rate


**GROUP 14**
MARA · Kolej MARA Kulim


**UN SDG 3**
Good Health and Well-being

Wellness tracker, not a medical device



# TRACK

FEATURES · THE HEALTH HALL

**WATER QUEST** Daily goal from your weight (35 mL/kg, the team's own rule). The knight walks, drinks and pours as you log; opt-in live weather adjusts your water potion.

**NUTRITION** Log from the Kolej MARA Kulim canteen menu. Signed in, **Luke** searches the web for any food, with a source for every part and an energy check (4·protein + 4·carb + 9·fat). Estimates are labelled.

**SLEEP** Hours against National Sleep Foundation ranges, a Sleep Score (our own game formula), Dream Battle, Counting Sheep and a wake-up alarm.

**STRESS QUEST** DASS-21 check-in (no severity labels), the Storm Within, Calming Grove breathing, five relief activities. Crisis numbers are always on screen.

**BODY & ENERGY** BMI (a screening number, not a diagnosis), BMR by Mifflin–St Jeor, daily energy = BMR × FAO/WHO/UNU activity level, macro and fibre goals.

**TRAINING HALL** 22 moves, 19 muscle maps; Calisthenics is bodyweight only. "I feel dizzy" and "Chest pain" stop buttons always at hand.

**RUNNING ROAD** GPS runs: distance, pace, calories (MET × kg × h, 2024 Compendium), opt-in Runners' Board.

**STAIRS & HEART RATE** BPM before, after and 1 minute after a climb for recovery. Camera pulse (video never stored) or a smartwatch / chest strap over Web Bluetooth.

**EVERYWHERE** Log for yesterday or any earlier date, browse every entry by day, print a weekly report.



# PLAY & GUIDE

FEATURES · RESTORE THE KINGDOM




**THE LIVING KINGDOM** Eight regions (Dream Realm, Stair Mountain, Energy Forge, Mind Forest, Heartstone Hall, Water Valley, Nutrition Village, Balance Tower), each in one of five states up to **Flourishing**.

**QUESTS & REWARDS** Adaptive daily and weekly quests, fair XP with a daily logging allowance, audited badges, gentle streaks, and mini-games for each log.

**HEALTH SCORE, 0-100** Five weighted indicators. The **Health Orb** turns it into ranked quest cards: Start Here, Next Step, Keep Going.

**MEDIUS NOTICES** Insights from your own baselines, a what-if simulator, week-on-week stats and the **Health Time Machine**.

**WIZARD'S COUNSEL** Talk with Medius or Luna (Gemini, through a secure server function). Signed-in only, rate-limited, with safety rules; nothing is stored. **Ask Luna** finds any feature without AI.

**ARCANA** The Oracle's crystal ball, Guild War of Joules and Dueling Arena: race a replay of your own earlier climb.

**OPT-IN COMMUNITY** Hall of Heroes (game progress only) and Runners' Board (nickname and weekly totals only).

**FOR EVERYONE** 8 themes, performance modes, reduced motion, keyboard and screen-reader friendly charts.



# PROVE IT

FEATURES · SCIENCE YOU CAN CHECK

**CAMPUS OBSERVATORY** A 3D map of Kolej MARA Kulim from OpenStreetMap, in Realistic and Fantasy looks. Each staircase is a heart coloured Mild, Moderate or Vigorous: tap to log a climb or workout. GPS position is never saved or sent.

**STAIR SENSE & SIGHT**

$h = n \times r$ $W = m \cdot g \cdot h$ $P = W \div t$

Camera and WebXR measure a real staircase's rise and tread.

**THE GROUP 14 STAIR EXPERIMENT**

Six participants climbed measured stairways. The pulse is modelled as **periodic motion** (f, T, ω) and $VO_2$ estimated with the project's method. Automated tests **recompute the report's tables from the raw counts**. If the pulse after is lower than before, the app says "not valid" instead of showing a number.

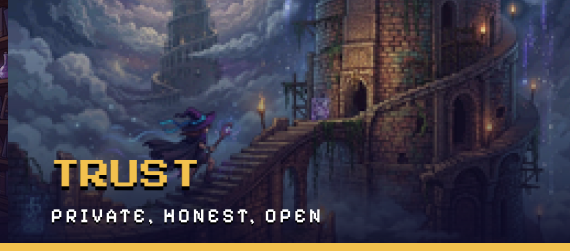
**EVIDENCE VAULT** Every source on a shelf: claim, how HealthWiz uses it, where you see it. Every "?" links to its entry.

**SCIENCE LABORATORY** Be Your Own Experiment, the Evidence Trial, and the **Uncertainty Engine** showing how unmeasured assumptions move an estimate.

**ASCENT MONUMENT** Your climbing builds an endless Tower of Babel, one stone per 0.25 m.

**ALCHEMIST'S WORKSHOP** Food energy, work against gravity and metabolic use are three different quantities; it shows why.

**Sources:** Mifflin–St Jeor · FAO/WHO/UNU · 2024 Compendium of Physical Activities · Keytel · Tanaka · Lovibond & Lovibond (DASS-21) · National Sleep Foundation · WHO · IOM. Where a rule is the team's own, the "?" says so.



# TRUST

PRIVATE, HONEST, OPEN

## YOUR DATA STAYS YOURS

- Saved **on your device**; download or restore a backup any time.
- A free account is **optional**: cloud backup and sync. Row Level Security means only you can read your row.
- **Delete account and cloud data** in two taps.
- Location only **when you tap**, never saved or sent. Camera pulse video never stored.
- AI chat is **never stored**; no AI key inside the app.

## UNDER THE HOOD

A static web app: plain JavaScript, no build step. Installable and offline-ready.

PWA · Supabase · Postgres + RLS · Edge Functions · Gemini · MapLibre · Three.js · WebXR · Web Bluetooth · Open-Meteo · Blender

## CREDITS AND LICENCES

Fonts: Press Start 2P and Pixelify Sans (SIL OFL 1.1). Map data © OpenStreetMap contributors (ODbL). MapLibre GL JS (BSD-3-Clause). Three.js (MIT). Map avatars: Tom, the anime school boy and girl (CC BY 4.0, suzuart). Campus look follows a drone video by @ikraMRahim, used with the creator's permission. Knight 3D base: MakeHuman CC0 assets. Emblems: MARA and Kolej MARA Kulim. All other characters and scenes are original HealthWiz designs.

## SAFE AND TESTED

Saved data is versioned and upgraded step by step, with a backup copy made before any upgrade or repair. **102** browser test files check every feature, with performance budgets and a device matrix. The **Complete Feature Atlas** lists every module.